

An Explainable AI (XAI) Model that Detects Pneumonia From X-Ray Images

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INTRODUCTION

Explainable Artificial Intelligence (XAI) refers to the practice of designing and developing AI models and systems in a way that their predictions, decisions, and behaviors are understandable and transparent to humans.

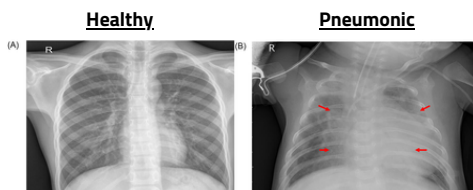
Transparency and Trust - XAI ensures that AI systems' actions are not perceived as "black boxes," but rather as mechanisms that can be comprehended and validated by users

Insight and Improvement - XAI aids in model improvement, data quality assessment, and the discovery of new knowledge.

Education and Adoption - XAI plays a crucial role in educating individuals about AI technologies

OBJECTIVES

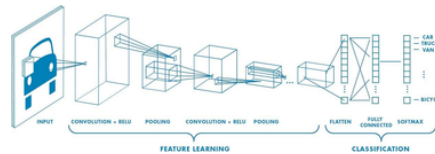
Our project aims to leverage eXplainable Artificial Intelligence (XAI) techniques to enhance the interpretability of Convolutional Neural Networks (CNNs) for the analysis of lung images, facilitating a more transparent and insightful examination of respiratory conditions.



The red arrows in **Figure B** indicate **white infiltrates**, a distinguishing feature of pneumonia

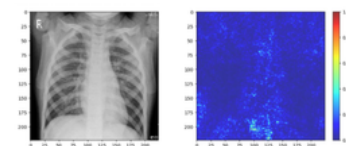
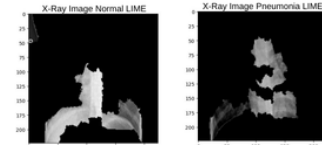
METHODOLOGY

- Created model using **Jupyter Notebook**
 - Trained on ~2000 X-Ray Images from Kaggle Dataset
 - Built using Python 3.10
- Libraries**
 - TensorFlow 2.10.0
 - Matplotlib & NumPy
- Convolutional Neural Network**
 - Step 1: Input Image
 - Step 2: Convolution Operation (with ReLu)
 - Step 3: Pooling
 - Step 4: Flattening
 - Step 5: Full Connection
 - Step 6: Output

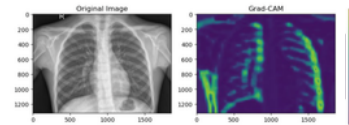


RESULTS

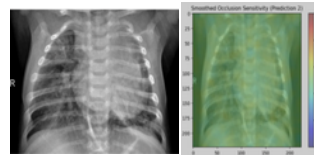
- LIME:** Local interpretation through perturbation-based explanations.
- Saliency Maps:** Identifying crucial image regions for model predictions.



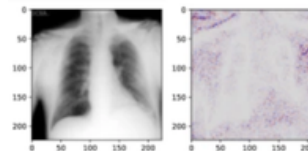
- Grad-CAM:** Visualizing CNN decision areas using gradient-based techniques.



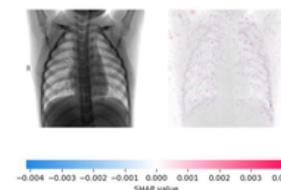
- Occlusion Tests:** Assessing feature importance via localized occlusion experiments.



- Deep Taylor Decomposition:** Decomposing model predictions for insightful feature attribution.



- SHAP:** Shapley values illuminate feature impact on predictions.



CONCLUSION

We successfully adapted established eXplainable Artificial Intelligence (XAI) methods to our models. Applying these XAI approaches to X-Ray Images enabled mostly successful differentiation between pneumonia-affected lung images and normal, healthy lung images. Our models offered a unique perspective by visually highlighting the specific areas within the images that influenced the Convolutional Neural Networks' decision-making process, shedding light on their inner workings.

FUTURE WORK

- Introduce diverse X-ray images to capture various scenarios and enhance model generalization.
- Investigate transfer learning using pre-trained models on larger medical image datasets.
- Fine-tune models on our specific pneumonia detection task for improved performance
- Perform systematic hyperparameter tuning to optimize model architectures
- Combine diverse models to mitigate individual model biases and enhance overall accuracy

ACKNOWLEDGEMENTS

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RESOURCES

Scan the QR Code to view the resources used in this project.

